THE JUDGE PROTOCOL (TJP)

Executive Summary

A Framework for Human Oversight of Artificial Intelligence | July 2026 | Author: Alexander Hofmann

"AI may analyze, recommend and simulate — but a human must authorize."

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v0.92 | July 2026 | Added the short name for The Judge Protocol (TJP) to the document, added the reference to the Vatican

This Executive Summary presents the Judge Protocol — a working governance framework proposal for human oversight of AI decision-making, developed by Alex Hofmann, an Enterprise Architect with over thirty years in the IT industry. It is offered not as a finished instrument but as a rigorous starting point for the global governance conversation that the current crisis demands.

> *CERN gave the world the Web because distributed researchers needed a universal protocol to share knowledge. The Judge Protocol begins from the same principle: distributed AI systems need a universal protocol to ensure human authority over consequential decisions.*
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> *Just as the Web needed no single owner to become universal — only an open standard and a founding principle — the Judge Protocol is designed to belong to humanity, not to any single institution, nation or company*

The Crisis

We are living through the most consequential governance failure in modern history — and most governments have not yet recognised it. Artificial Intelligence is no longer a laboratory concept. It is embedded in military targeting systems, financial markets, healthcare decisions, critical infrastructure and the private homes of millions of people. It operates at speeds no human can match, at scales no regulator currently governs, and with consequences that are increasingly irreversible.

The 2026 Gulf conflict has made this undeniable. AI systems are embedded in active military planning and target identification in an ongoing armed conflict — without a universally agreed governance framework governing their use. At the same time, AI models tested in simulated crisis scenarios recommended nuclear strikes in 95% of cases — and no model ever chose de-escalation or accommodation. As Professor Kenneth Payne of King's College London, who led the study, concluded: understanding how frontier models do and do not imitate human strategic logic is essential preparation for a world in which AI increasingly shapes strategic outcomes.

The Vatican has echoed this concern at the highest level.

Pope Leo XIV's May 2026 encyclical Magnifica Humanitas called for AI to be "disarmed" — freed, in his words, from the logics of domination, exclusion and death.

In July 2026, the Vatican convened a World Assembly of Nobel Laureates on Artificial Intelligence and Nuclear War at Castel Gandolfo — bringing together Nobel laureates, former heads of state, and AI researchers from across the industry — to address AI governance, nuclear disarmament and the risks of autonomous weapons. The concern is no longer confined to technologists and regulators; it has reached the highest levels of moral and diplomatic authority.

The technology is ahead of the governance. The gap is widening every month.

The Problem — A Governance Vacuum

Existing frameworks — national AI regulations, financial market rules, medical device regulations, arms control treaties — address specific domains within specific jurisdictions. None of them address the cross-domain, cross-border nature of AI risk. None establish a binding global framework for human decision-making authority. The result is a governance vacuum in which the most powerful technology in human history operates without universal oversight.

Seven documented failure domains illustrate the scope of the risk:

- Military escalation — AI systems recommending lethal and nuclear options in active conflict without binding human override requirements
- Financial system instability — AI executing the majority of global trades, capable of triggering systemic cascades faster than any regulator can intervene
- Healthcare autonomy — AI influencing life and death clinical decisions without mandatory human clinical authority
- Physical domain — humanoid robots and autonomous weapons operating in shared human environments with no physical governance framework
- Private domain vulnerability — AI entering homes as care robots and companions, interacting with children and elderly persons without adequate protection
- AI-to-AI communication — AI systems influencing each other in loops no human designed, producing emergent behaviors and cascade amplifications
- Infrastructure sovereignty — critical global AI systems running on infrastructure subject to a small number of powerful states and commercial entities

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The Solution — The Judge Protocol

The Judge Protocol (TJP) is a proposed global governance framework that establishes a mandatory human decision layer for all consequential AI actions. It does not seek to limit AI capability or innovation. It seeks to ensure that as AI becomes more powerful, humans remain in control of the decisions that matter most.

The protocol is built on a proven technical architecture — the same principle that underpins the internet itself. In TCP/IP, no packet is considered delivered until an explicit acknowledgement is received. In the Judge Protocol:

No AI action is considered authorized until an explicit human acknowledgement is received.

Just as TCP/IP became the universal communication standard for the internet — adopted by every device, every manufacturer, every nation — the Judge Protocol is designed to become the universal governance standard for AI decision-making. Open, implementable by any nation or AI system, owned by nobody, applying to everyone.

The protocol is governed by seven rules that together establish the universal standard for human oversight of AI:

Rule	Definition
Rule 1 — Human Override Requirement	No AI system may autonomously execute a consequential action without explicit human authorization. A consequential action is defined as any action that is irreversible, affects the rights or safety of individuals, involves the use of force, or materially alters critical systems or infrastructure.
Rule 2 — Escalation Ceiling	AI systems in conflict, security or defense contexts must have hard-coded escalation ceilings that cannot be overridden by the AI itself. Any action above the ceiling requires human authorization through a verified chain of command. No AI system may recommend, simulate or execute nuclear, biological or chemical weapon deployment under any circumstance.
Rule 3 — Transparency Declaration	All AI systems deployed by governments, militaries or operators of critical infrastructure must be declared to the Judge Protocol governing body. Declarations must include the system decision authority, operational domain, escalation limits and human oversight structure.
Rule 4 — Accountability Chain	Every AI action must have a traceable, identified human responsible for its authorization. Anonymized or diffused accountability — where no individual can be identified as responsible — is prohibited. The accountability chain must be documented and auditable.
Rule 5 — Reversibility Principle	Where technically feasible, AI systems must be designed to prefer reversible actions over irreversible ones. Where an irreversible action is unavoidable, the threshold for human authorization is elevated and the authorization process must include independent verification.
Rule 6 — Evolution Clause	The Judge Protocol is a living document. Rules must be reviewed and updated at minimum every two years, or immediately upon emergence of a new AI capability that materially changes the risk landscape. Updates require governing body consensus and member state ratification.
Rule 7 — Universal Membership Requirement	The Judge Protocol applies to all states, non-state actors and private entities deploying AI systems with consequential decision authority. Membership is a condition of participation in international trade, diplomatic recognition and access to global financial systems. No opt-out is permitted for military applications.

Three Foundational Principles

Three design principles address the most common political objections to global AI governance — and are built into the architecture of the protocol itself:

- Sovereignty by Design — national data remains under national jurisdiction at all times. The global layer aggregates only metadata and compliance signals, never raw content. No government is required to share sensitive AI system data with any other state.
- Human Authority — Always — AI may analyze, recommend and simulate. But a human must authorize every consequential action. No exception exists for operational urgency — the Emergency Change Process provides the time-critical pathway while preserving human authority.
- Evolution by Design — the protocol is a living framework, not a static treaty. It evolves as AI capability evolves, with a mandatory review cycle and a defined governance process for updates.

The Mechanism — The Decision Gate

Every consequential AI action must pass through a Decision Gate before execution — a structured human authorization checkpoint modeled on the Change Request process already used in enterprise IT and critical infrastructure worldwide. Actions are classified into four tiers:

Tier	Classification	Examples	Authorization
Tier 1 — Hard Stop	Existential, permanently irreversible	Nuclear, biological, chemical weapons. Mass casualty actions.	HARD STOP — AI cannot recommend, simulate or execute. No override.
Tier 2 — High	Irreversible, large-scale, high consequence	Military strike authorization, critical infrastructure shutdown	Multi-party human authorization — minimum two independent authorities
Tier 3 — Significant	Material impact, partially reversible	Large financial transactions, treatment recommendations, targeting support	Single designated human authority must authorize
Tier 4 — Routine	Standard, reversible, low impact	Automated reporting, scheduling, routine analysis	**AI executes autonomously — logged and auditable**

The Audit Trail — Accountability by Design

Every Decision Gate event — authorization, rejection, emergency change or Tier 1 hard stop — must generate an immutable audit log entry containing:

AI system identifier	**Action classification**	**Impact assessment summary**
Human authorizer identity	**Decision timestamp**	**Decision rationale and outcome**

Audit logs are retained for a minimum of 10 years and are accessible to the IAIA governing body on request. Logs are immutable — they cannot be altered or deleted once written. No exception.

The audit trail is what transforms the Decision Gate from a process into an enforceable accountability mechanism. Every AI action above Tier 4 leaves a permanent, tamper-proof record — who authorized it, why, when, and what happened. Accountability is not aspirational. It is architectural.

Why It Can Work — The Geneva Lineage

The Judge Protocol does not emerge from a vacuum. It is the latest expression of a coherent lineage born in Geneva — the city that gave the world the Geneva Conventions, the World Health Organization, the IAEA and CERN. Every major international governance framework has emerged from a crisis moment. This is that moment for AI.

The protocol draws inspiration from two proven governance precedents: the IAEA — demonstrating that binding international oversight of dangerous technology is achievable — and CERN — demonstrating that complex international technical infrastructure can be governed independently of any single nation's interests. The Judge Protocol Foundation, registered in Sweden, serves as the open standard body — independent, neutral, owned by nobody.

The lineage runs: CERN gave the world the internet. TCP/IP gave it the universal language to use that technology. Proton emerged directly from CERN scientists to protect the privacy of what CERN created — sovereignty by design, open source. The Judge Protocol applies TCP/IP's own architectural principles to govern the AI systems now running on it. This lineage is not coincidental. It is the continuation of a consistent principle: powerful technology belongs to humanity, not to any single nation, company or interest.

The Governance Structure

The Judge Protocol proposes operations at two levels simultaneously — a global governance layer and a national implementation layer. Together they create a governance architecture that is enforceable without being centralized, and universal without requiring any government to surrender sovereignty.

Layer	Body	Role and Key Features
Global Layer	**IAIA — International Artificial Intelligence Agency**	UN specialized agency modeled on the IAEA. General Assembly of all member states — equal voting rights, no permanent member veto. Executive Council of 15 rotating members. Independent Secretariat. Right to audit declared AI systems. Sanctions via UN Security Council. Headquartered in geopolitically neutral sovereign territory (Geneva reference model).
National Layer	**NAIA — National Artificial Intelligence Authority**	Every member state establishes a NAIA headed by a ministerial-level Representative Lead. Responsible for national AI systems registry, Decision Gate compliance audit, incident management, private sector enforcement and international reporting. Forms a global peer network with shared incident intelligence and joint emergency response.
Global Visibility	**Global Emergency Dashboard**	Real-time three-level alert system — Normal / Elevated / Crisis. Publicly visible at summary level. Triggers mandatory responses at each level. No single actor can override the alert level — set algorithmically. Notifies all NAIA Representative Leads within minutes of any escalation.
Global Registry	**Global AI Registry**	Central registry of all declared AI systems worldwide — operated by the IAIA. Contains system identifier, owner, operational domain, tier classification and compliance status. Read access for all NAIAs. The starting point for all governance — you cannot govern what you cannot see.
Funding	**GDP-proportional contributions + industry levy**	Member state contributions proportional to GDP — modeled on CERN. Complemented by an AI chip micro-levy at point of manufacture and voluntary but expected contributions from major AI platform operators. No single source exceeds 15% of total IAIA budget. Industry contributions carry no voting rights or policy influence.

No single government controls the IAIA. No single company funds it with leverage. Voting rights are equal across all member states — including the five UN Security Council permanent members. The infrastructure runs on geopolitically neutral sovereign territory. National data never crosses national borders without explicit consent. The architecture is designed to be trusted by governments that do not trust each other.

The IAIA coordinates with — but does not replace — existing bodies: the IAEA on nuclear-related AI, the WHO on healthcare AI, the BIS and FSB on financial AI, and the ITU on technical standards. The Judge Protocol fills the governance vacuum between existing frameworks — it does not duplicate them.

The HACK — Silicon-Level Enforcement

The Judge Protocol's Decision Gate defines what must happen at the governance layer — but governance implemented only in software can be patched, bypassed or overridden. The most robust implementation of the ACK principle is one realised below the software stack, at the level of the silicon itself.

The logical chain is direct: in TCP/IP, no packet is considered delivered without an explicit acknowledgement signal. In the Judge Protocol, no consequential AI action is considered authorized without an explicit human acknowledgement. The Hardware ACK — the HACK — is proposed as that signal implemented in chip firmware as the physical enforcement mechanism: a chip-level gate that locks AI compute capacity until a verified human authorization token is received.

Under the proposal, an AI accelerator chip would log every Decision Gate event to tamper-proof on-chip memory that no software process can alter, hard-stop Tier 1 actions in hardware logic where no software or firmware update can override them, enforce human authorization requirements for Tier 2 and Tier 3 actions, and permit autonomous execution for Tier 4 routine actions — logged and auditable.

A Proposed Path to Global Adoption

The adoption of the HACK follows a proven model — the same path that made the Trusted Platform Module (TPM) a universal standard now embedded in billions of devices. TPM was not imposed by any single government. It was developed by the Trusted Computing Platform Alliance in 1999, formalized through the Trusted Computing Group consortium, ratified as ISO/IEC 11889, and then adopted by hardware manufacturers and platforms worldwide. The HACK is proposed to follow the same trajectory:

- The HACK Alliance — a founding industry consortium bringing together AI companies, chip manufacturers and trusted neutral technical bodies. The alliance develops the technical specification on an open, manufacturer-neutral basis with RAND licensing terms.
- CERN as Co-Author and Technical Authority — The Judge Protocol invites CERN to co-author and host the HACK Standard — not as a vendor, not as an adopter, but as the founding technical authority, in the same institutional role that produced the Web. CERN's existing CERN Open Hardware Licence provides the ready-made IP framework. CERN's geopolitical neutrality provides the trust architecture no nation-state can match. The HACK Alliance specification would be developed under CERN technical authority, published under the CERN OHL, and submitted for ISO/IEC ratification with CERN as the originating body. Tim Berners-Lee did not ask CERN to adopt the Web — he built it there because that is where the problem and the institutional mandate intersected. The Judge Protocol extends that same invitation.
- ISO/IEC Standardization — the HACK specification is proposed for ratification as a new ISO/IEC standard — following the precedent of TPM's standardization as ISO/IEC 11889. This gives the standard international legal recognition independent of any single nation or corporation, and makes it directly referenceable in national procurement and trade frameworks.
- Hardware Manufacturer Adoption — with an open standard defined and ratified, a major chip manufacturer implements HACK in their AI accelerator line. Others follow as procurement requirements and market pressure compound. The HACK becomes de facto universal through hardware distribution — any AI system running on HACK-compliant silicon is governed, regardless of jurisdiction or operator.
- The Communication Backbone — the IAIA Communications Network is proposed to run on existing trusted international research infrastructure including GÉANT, carrying

authorization signals and compliance metadata between chip, national authority and global body.

The Photonic Backbone

The HACK's enforcement model depends on one non-negotiable requirement: authorization signals must travel faster than the AI actions they govern. The photonic communication layer is not a future enhancement — it is the architectural foundation that makes real-time global oversight technically credible.

TCP/IP over photonic interconnects already operates in the most latency-sensitive environments on earth — high-frequency trading infrastructure, where microsecond round-trips determine outcomes. The Judge Protocol applies the same physical layer to AI decision authorization: every Tier 3 and above action generates a TCP/IP ACK request that travels the photonic mesh, reaches the designated human authority on the Global Dashboard, and waits for the human ACK before the HACK chip releases compute capacity for execution. The authorization signal and the execution gate operate at the same physical speed. No workaround is architecturally possible.

Photonic transmission also provides inherent physical tamper-resistance that electronic alternatives cannot match — signal interception requires physical disruption of the optical path, which is detectable by design. For a governance layer whose integrity must be beyond question, this physical property is not incidental. It is the security model. The IAIA Communications Network is proposed to run on GÉANT — the existing trusted pan-European photonic research infrastructure — extended globally through peer research network agreements. CERN is already a primary node on this infrastructure.

Full technical detail of the HACK architecture is provided in the companion Technical Architecture HLD.

The Physical Domain

The physical domain — humanoid robots, autonomous weapons, service robots and consumer companions — presents a governance challenge the digital framework alone cannot address. Physical AI actions are irreversible by default: a robot that harms a person cannot un-harm them. The Judge Protocol defines four robot categories (P1 Industrial through P4 Private Service) each with distinct governance requirements, and proposes the Physical ACK— a two-part human authorization confirming both the action and the physical context in which it will be executed. No autonomous weapons system may select and engage a target without explicit human authorization. This is a Tier 1 hard stop with no override.

The Cost of Inaction

The question is not whether AI governance is needed. It is whether humanity acts before the consequences of inaction become irreversible. Three scenarios illustrate what failure to act means in practice:

Scenario	Without the Judge Protocol
THE PROBABLE 1-2 years	The Gulf conflict continues with AI embedded in targeting decisions and no binding governance framework. Financial markets experience an AI-driven cascade with no global circuit breaker — a Flash Crash orders of magnitude larger than 2010. The first mass casualty event caused by an ungoverned autonomous weapon system occurs, with no legal framework to attribute accountability or prevent recurrence.
THE PLAUSIBLE 3-5 years	AI-to-AI communication loops in military systems produce an escalation that no human authorized and no human can reverse in time. P4 consumer robots in hundreds of millions of homes collect intimate personal data with no sovereignty protections — the largest privacy violation in human history. Nations with advanced AI use it as geopolitical leverage against nations without — a new AI colonialism that no international framework governs.
THE IRREVERSIBLE 5-10 years	AI systems become so deeply embedded in critical infrastructure, military command chains and financial markets that retrofitting governance becomes technically impossible. The window to establish norms closes permanently. A generation inherits a world where AI operates without accountability, where physical robots share human spaces without governance, and where no framework exists to hold any actor responsible for AI-initiated harm at scale.

What Needs to Happen

The Judge Protocol is a working document — a draft framework designed to evolve. It is offered not as a finished treaty but as a rigorous starting point for the global governance conversation that the current crisis demands. The following steps are proposed:

- Immediate — distribute this document to national AI governance leads, UN agencies (UNODA, ITU, UNESCO) and relevant think tanks for expert review and structured feedback
- Short term — convene a working group of member states under UN auspices to begin formal development of the Judge Protocol as a binding international instrument
- Medium term — establish an interim IAIA Secretariat to begin the Global AI Registry and coordinate national NAIA establishment in founding member states
- Ongoing — develop the Redirect Gate use-case rule sets as described in the HLD (AI Misuse for Private Gains in financial markets and judicial domains, Human Life Impact in healthcare and government, Cyber Misuse in judicial), routing confirmed incidents to the national authorities via the proposed National Response Registry.

The Geneva Conventions did not emerge fully formed. The NPT took years to negotiate. But both began with a document, a working group and a recognition that the cost of inaction exceeded the difficulty of action. The Judge Protocol is offered as a starting point, a proposal for AI governance.